

Multiple Choice (10 marks)

1. What hormone is responsible for metamorphosis in amphibians?
 - (a) growth hormone
 - (b) estrogen
 - (c) thyroxine
 - (d) insulin
 - (e) prolactin
2. The ability of the brain to detect differences in stimulus intensity is best explained by the fact that which of the following varies with the stimulus intensity?
 - (a) The number of synapses crossed
 - (b) The speed of the action potential
 - (c) The amplitude of the action potential
 - (d) The number of neurons activated
 - (e) The refractory period of the neuron
3. Two individuals, one with type B blood and one with type AB blood, have a child. The probability that the child has type O blood is
 - (a) 50%
 - (b) 100%
 - (c) 0%
 - (d) 10%
 - (e) 75%
4. Keystone species are thought to have profound effects on the structure and composition of ecological communities because they
 - (a) are more abundant than most other species in their communities
 - (b) tend to reduce diversity by eliminating food resources for other species
 - (c) are the largest species in their communities
 - (d) are the most aggressive species in their communities
 - (e) are the only species capable of reproduction
5. A deficiency in which endocrine gland would result in tetany?
 - (a) Pancreas

- (b) Thymus
- (c) pituitary gland
- (d) adrenal gland
- (e) thyroid gland

True / False (10 marks)

Answer True or False

6. True or False: All tissue types in higher plants have identical functional roles, making classification straightforward and simple.
7. True or False: Organ transplants are generally unsuccessful due to a lack of available organs for patients in need.
8. True or False: The pH of both lakes will decrease in response to acid rain, indicating increased hydrogen ion concentration.
9. True or False: A single nucleotide insertion after the first codon position is less likely to disrupt protein function than a deletion of a nucleotide triplet.
10. True or False: Mules can reproduce asexually, allowing them to pass on their genetic traits to offspring.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the removal of product C from the equilibrium reaction $A + B \rightarrow C$, catalyzed by enzyme K, can influence the production of more product C. Include the principles of Le Chatelier's principle in your response.
12. Discuss how a cross between two brown-furred minks can produce silver-furred offspring. Include a detailed analysis of the genotypes involved and the principles of inheritance at play.

End of Paper